Jack Lundholm

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Education

Engineering Physics Aug 2025 - Present

Bachelor of Applied Science in Engineering, University of British Columbia - Dean's List *Vancouver, BC*

Skills and Certifications

Skills: Autocad, Plant 3D, Solidworks, C, Java, Python, MATLAB, Solidworks Fluid Dynamics, Arduino Inventor, Arduino, SewerGEMS, WaterGEMS, Altium, nRFT Firmware dev, Microsoft Project.
Awards/Certifications:UBC Project Shop WHMIS (2026), Halcyon Days (2025), Fermat Math Contest School Winner (2024), TDSS Physics, Math and Chemistry Awards (2025)

Profile

Second-year UBC Engineering Physics student with experience designing and implementing control systems, electromechanical mechanisms and back-end pogram development. Proficient in a variety of CAD software, with demonstrated ability to learn new technologies quickly and contribute to project success. Proven ability to troubleshoot complex technical challenges and collaborate effectively within multidisciplinary engineering teams.

Technical Experience

Mechanical Engineering Summer Student May–Aug 2025, May–Aug 2026

EXP *New Liskeard Office, ON*

- Designed and drafted engineering deliverables for 20+ contracted projects, including a water treatment plant, sewer and water distribution system modeling, lift station design, and process infrastructure for cobalt sulphate and nickel processing facilities.
- Developed client-facing design briefs for water/sewer system modeling, fire prevention studies, and lift station projects.
- Validated design accuracy through drawing reviews with engineers and incorporated client feedback gathered during project instruction meetings.
- Automated file management workflows for the office's largest active project by programming a custom Python tool, reducing estimated administrative processing time by 30%.
- Improved team efficiency by training coworkers on industry-standard software, including SolidWorks, SewerGEMS, WaterGEMS, Plant 3D, AutoCAD, Inventor and Python.

Elec + Firmware and Propulsion Team Member | *SUBC Design Team* Sept 2025 - Present

- Member of submarine design team placing 8th at the eISR 2026 international competition.
- Designed control boards and wrote firmware for said boards in implementing an electric steering sytem into the submarine.
- Engineered data collection firmware for the harvesting of data from sensors including hall effect sensor, barometer and more using the nRF connect SDK.
- Designed, fabricated and tested 4 different propellor prototypes using multitude of CAD softwares for submarine in consideration of both performance and manufacturing.
- Conducted research and preliminary designs for future propulsion solutions, (single bladed prop, biomimetics), and presented the findings.

Extended Learning Experience

PCB Designer | *PCB Design and Prototyping ELab* Sept 2026 - Nov 2026

- Developed PCB circuit layouts and planning utilizing KiCad software.
- Optimized layouts for minimizing noise level with grounding layout techniques.
- Developed strategies for digital, analog, and mixed signal PCB design
- Manufactured PCB design layouts to generate a final product.